

|       |          |                  |                    |       |
|-------|----------|------------------|--------------------|-------|
| No    | No       | No               | Commission         | Lower |
| Route | Slippage | Impermanent loss | Allocation by Role | Gas   |

# ***TTSWAP***

Detailed Design Whitepaper based on Market  
Value Transaction Conservation Principle

2024-04-01

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# 1 Summary

TTSWAP (token-token swap) is an automated market-making protocol built on the EVM blockchain, which means it doesn't rely on centralized institutions or individuals to facilitate trades. The core principle of TTSWAP is to automatically trigger market value transfers based on user actions, creating a platform based on a constant value trading model.

The project's whitepaper explains the design logic of TTSWAP, covering the following aspects:

1. ToKen Trading:

Users can directly swap one token for another without the need for intermediary tokens.

2. Value Token Investment and Withdrawal:

Users can invest in specific value tokens and withdraw their investments when needed.

3. Ordinary Token Investment and Withdrawal:

Besides value tokens, users can also invest in ordinary tokens and withdraw their investments at any time.

4. Generation and Distribution of Transaction Fees:

The transaction fees generated during trades are distributed according to certain rules to incentivize more participants to join the market.

In summary, TTSWAP provides ordinary users with a simple, transparent, and efficient cryptocurrency trading platform that uses an innovative AMM logic—the constant value trading model. It aims to create a convenient, secure, and low-GAS fee platform.

## 2 Features

1. Value conservation trading strategy

The value conservation trading strategy accurately reflects the true market value of currencies and facilitates fast token transactions.

2. Direct trading without intermediaries

On this platform, any two types of items can be directly traded without the need for intermediate conversions.

3. No slippage within trading threshold

Transactions below the token trading threshold incur no slippage.

4. No impermanent loss for liquidity providers or token investors

Constant market value inherently prevents impermanent loss. When users withdraw their investment, they receive the original invested token plus profits generated from providing liquidity.

5. Low gas fees with simple computational logic

The logic behind the constant value trading model is simple, resulting in low computational load and gas consumption.

6. Fee distribution based on roles for everyone

Fees are distributed based on roles, allowing anyone to become a token investor (liquidity provider), token manager, portal, referrer, user, or platform role, sharing in the platform's growth earnings.

7. Support native ETH exchange and invest

anyone can you native ETH without wrap to swap, invest easily.

## 3 Principle of Value Conservation Trading Mode

### 3.1 Value Conservation Trading Mode

$$\frac{V_A}{Q_A} * \Delta A = \frac{V_B}{Q_B} * \Delta B = \dots = \frac{V_Z}{Q_Z} * \Delta Z \quad (1)$$

$$V_A : \text{Record tokenA's market value} \quad (2)$$

$$Q_A : \text{Record tokenA's quantity} \quad (3)$$

$$\Delta A : \text{Record tokenA's change amount(buy or sell)} \quad (4)$$

$$V_B : \text{Record tokenB's market value} \quad (5)$$

$$Q_B : \text{Record tokenB's quantity} \quad (6)$$

$$\Delta B : \text{Record tokenB's change amount(buy or sell)} \quad (7)$$

$$V_Z : \text{Record tokenZ's market value} \quad (8)$$

$$Q_Z : \text{Record tokenZ's quantity} \quad (9)$$

$$\Delta Z : \text{Record tokenZ's change amount(buy or sell)} \quad (10)$$

Market value is a measure of the level of demand from users for the tokens in the protocol. When a user sells tokens, it indicates a decrease in demand for the tokens in the agreement, resulting in a

decrease in the market value of that product. When a user purchases a token, it indicates an increase in demand for the agreed quantity of tokens, resulting in an increase in the market value of that tokens

### 3.1.1 Compute Logical

- A,B's status before trading

$$A's \text{ market value in protocol} : V_A \quad (11)$$

$$A's \text{ quantity in protocol} : Q_A \quad (12)$$

$$B's \text{ market value in protocol} : V_B \quad (13)$$

$$B's \text{ quantity in protocol} : Q_B \quad (14)$$

alter trading  $\Delta A$  for  $\Delta B$

$$A's \text{ market value in protocol} : V_A - \frac{V_A}{Q_A} * \Delta A \quad (15)$$

$$A's \text{ quantity in protocol} : Q_A + \Delta A \quad (16)$$

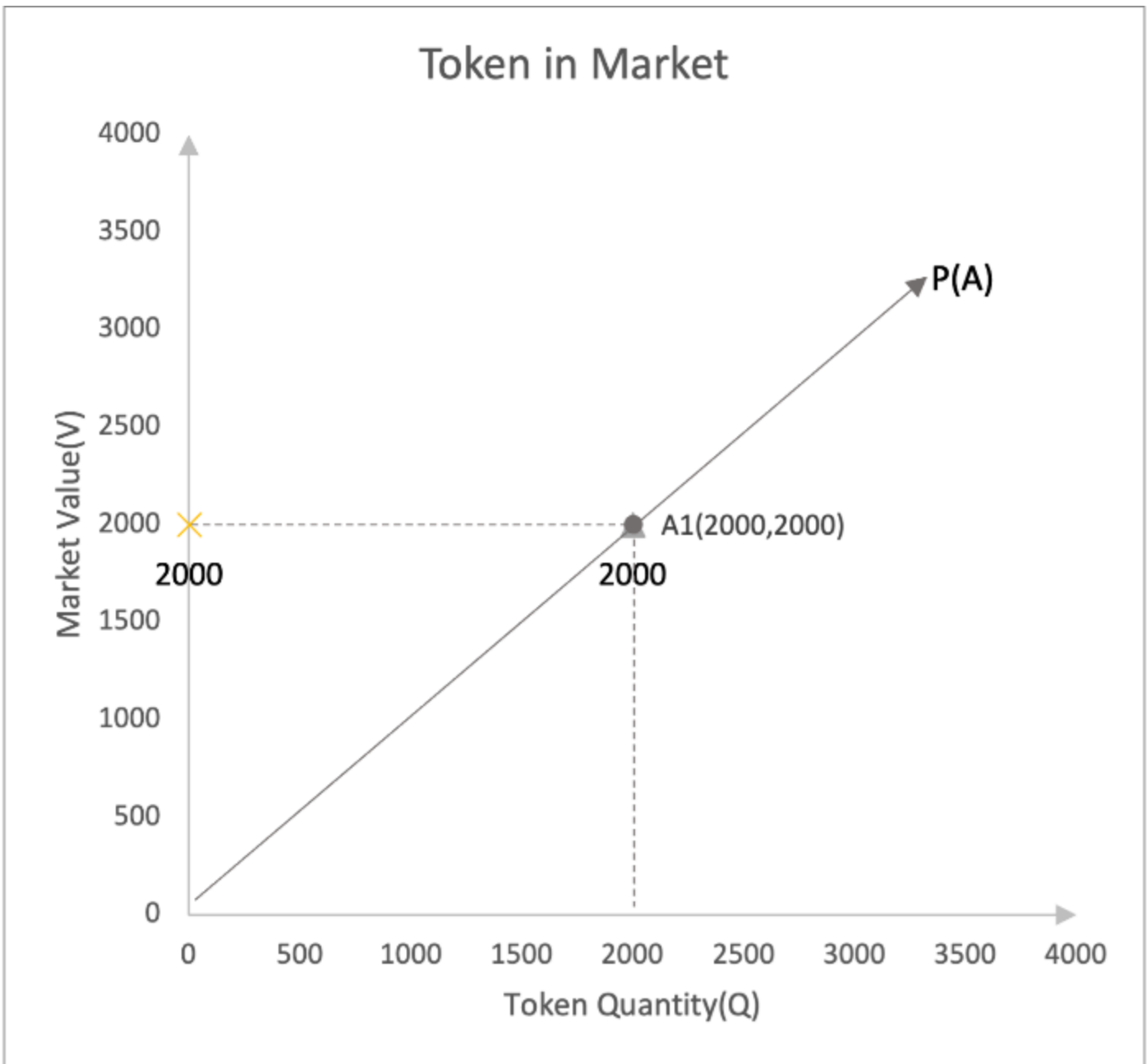
$$B's \text{ market value in protocol} : V_B + \frac{V_A}{Q_A} * \Delta A \quad (17)$$

$$B's \text{ quantity in protocol} : Q_B - \frac{\frac{V_A}{Q_A} * \Delta A * Q_B}{V_B} \quad (18)$$

$$\text{receive } \Delta B = \frac{\frac{V_A}{Q_A} * \Delta A * Q_B}{V_B} \quad (19)$$

## 3.2 Tokens

Example: There are 2000 units of token A1 in the market, with a market value of 2000.



Definition:

Market value  $V(A1)$ : Used to measure the degree of demand for tokens in the market. The market value of token A is 2000.

Quantity  $Q(A1)$ : Records the quantity of tokens. The quantity of token A is 2000.

Unit value  $P(A1)$ : The market value per unit quantity. The unit value of token A is 1.

## 3.2 The Relationship between Market tokens and User Buying and Selling Behavior

- Example 1: The user spends a value of 1000 to purchase token A.  
The demand for token A in the market increases.  $V(A1) = 2000 + 1000 = 3000$ .

The quantity of token A in the market decreases.  $Q(A1) = 2000 - 1000 = 1000$ .

The unit value of token A in the market changes.  $P(A1) = 3$ .

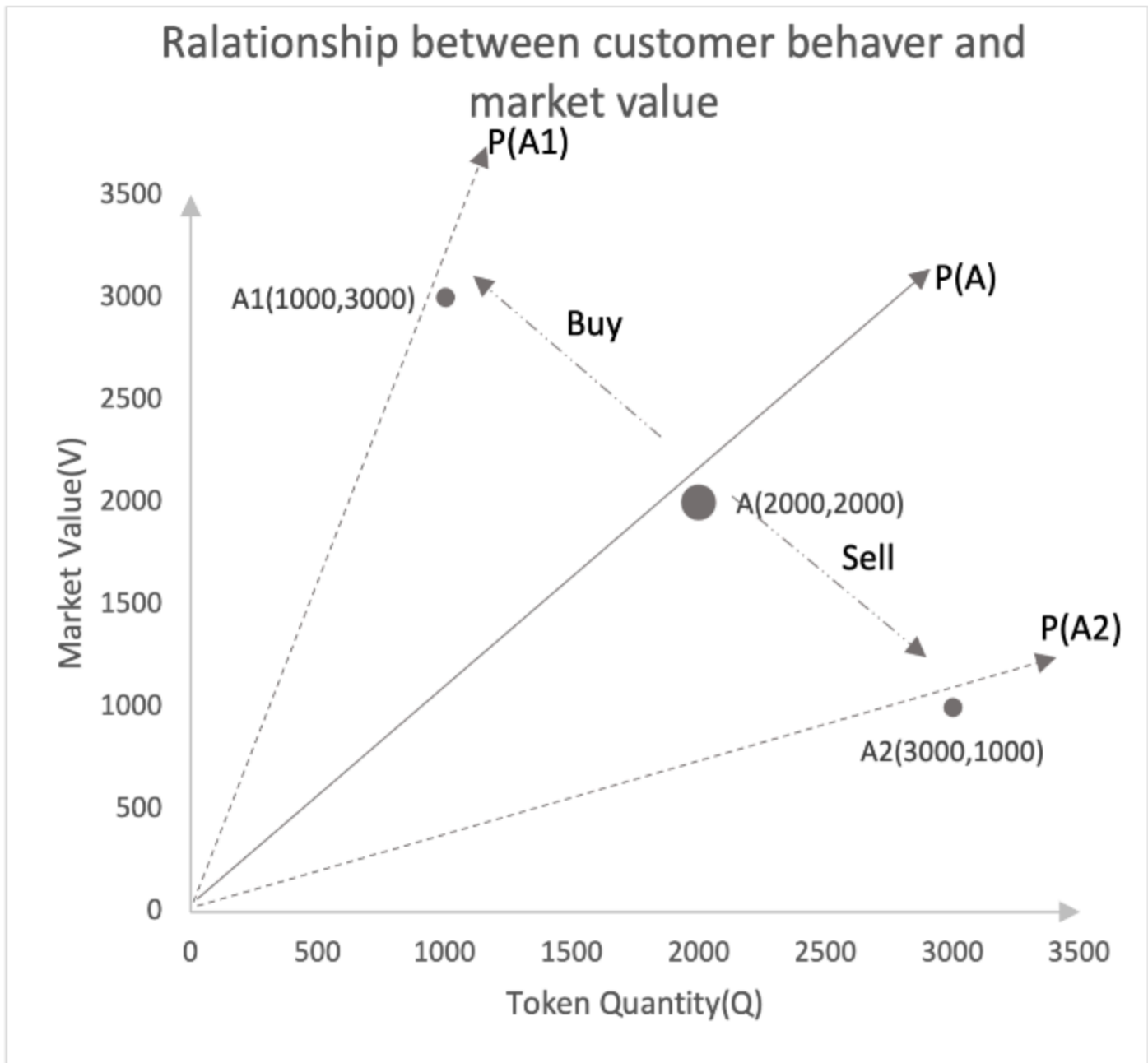
- Example 2: The user sells token A for a value of 1000.

The demand for token A in the market decreases.  $V(A2) = 2000 - 1000 = 1000$ .

The quantity of token A in the market increases.  $Q(A2) = 2000 + 1000 = 3000$ .

The unit value of token A in the market changes.  $P(A2) = 0.3333$ .

Display as shown in the following diagram



### 3.3 The relationship between user behavior and token status in the market

Now, as users sell and purchase, the market value  $V$  and quantity  $Q$  of tokens change, causing corresponding changes in the token prices. The changes in market value  $V$  and quantity  $Q$  of tokens are depicted in the graph

Alt text

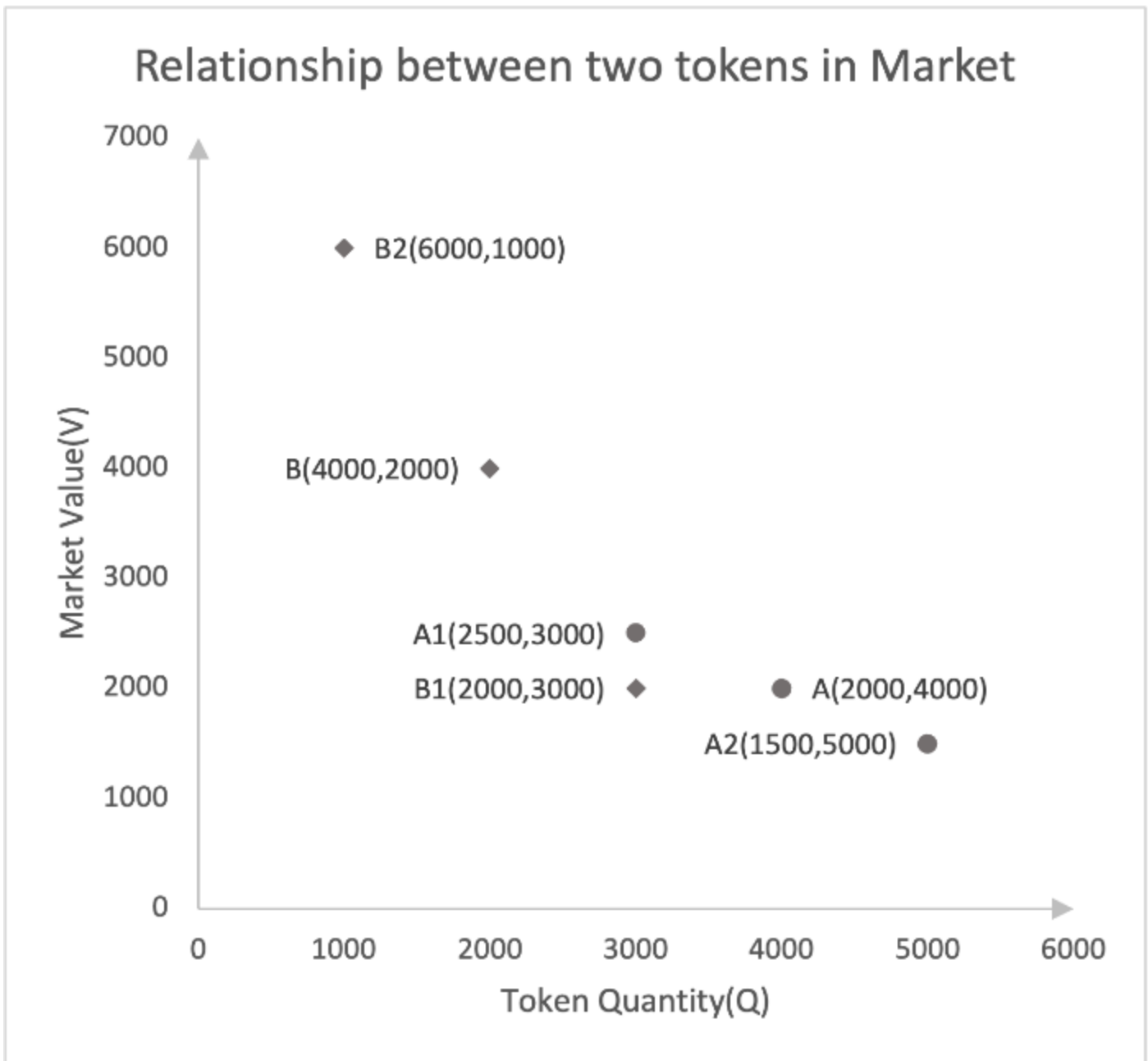
### 3.4 The relationship between two types of items in the market

Now in the market, there are two types of tokens, A and B.  $A(2000, 4000)$ ,  $B(4000, 2000)$ .

When users use 500 units of token A, the corresponding market value is 1000. The market value of 1000 corresponds to 1000 units of token B.

When users use 500 units of token A, they can purchase 1000 units of token B. In the graph below, A will move to position A1, and B will move to position B1.

When users sell 500 units of token A, they will obtain 1000 units of token B. In the graph below, A will move to position A2, and B will move to position B2.

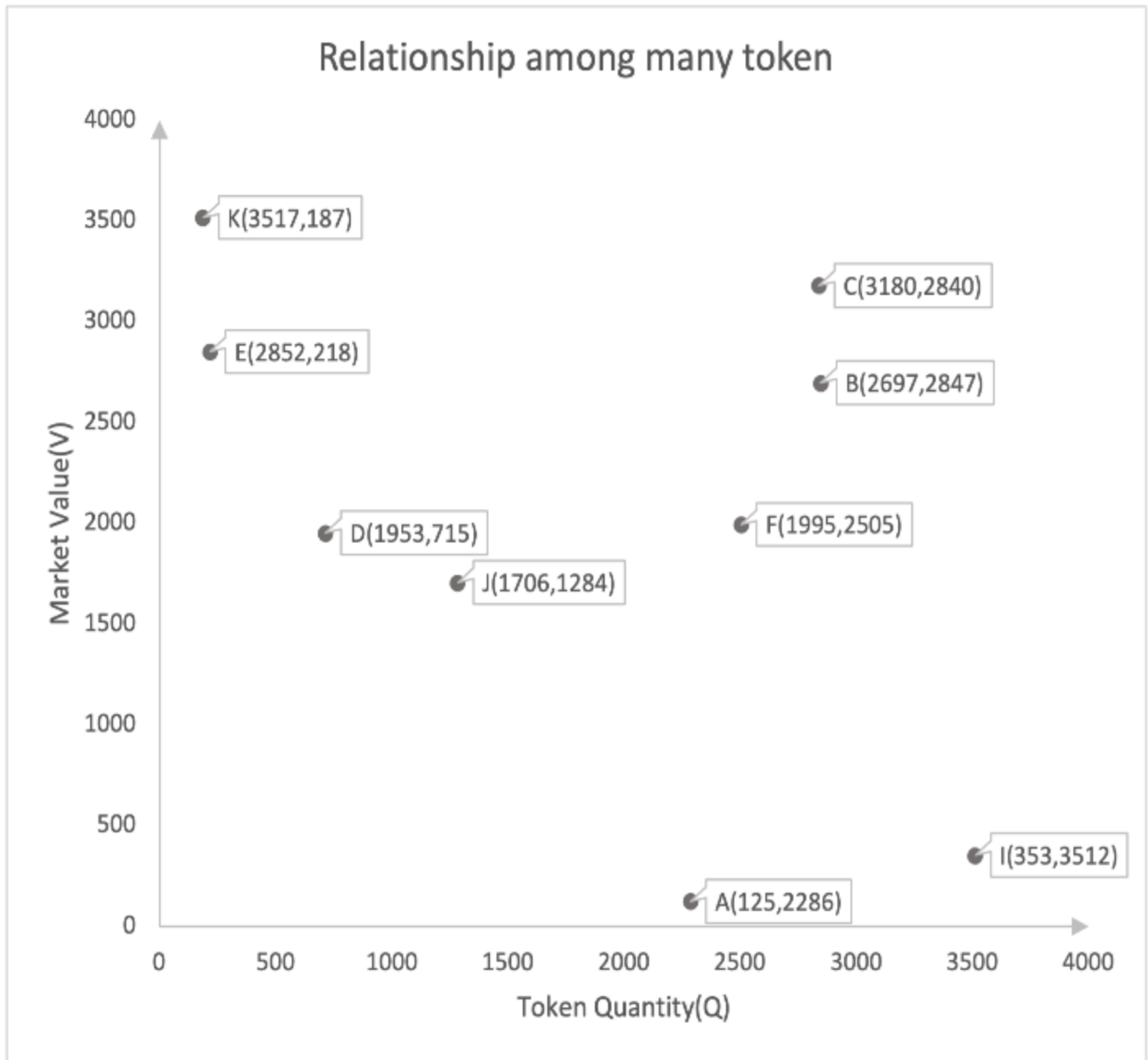


Due to the change in position,  $P(A)$  and  $P(B)$  also change. The relative price of token B to token A will also change. If there is a discrepancy with the external market price, other transactions will facilitate the convergence of market price with the external market price.

Note: If the proportion of purchase quantity in the market data is too large, it will cause significant fluctuations in the relative prices of the two tokens. Therefore, each transaction will be split into multiple smaller orders for trading.

### 3.5 The relationship between multiple tokens in the market

User transactions cause changes in the positions of any two tokens, which in turn affect the positions of these two tokens relative to other tokens, resulting in synchronized price changes.



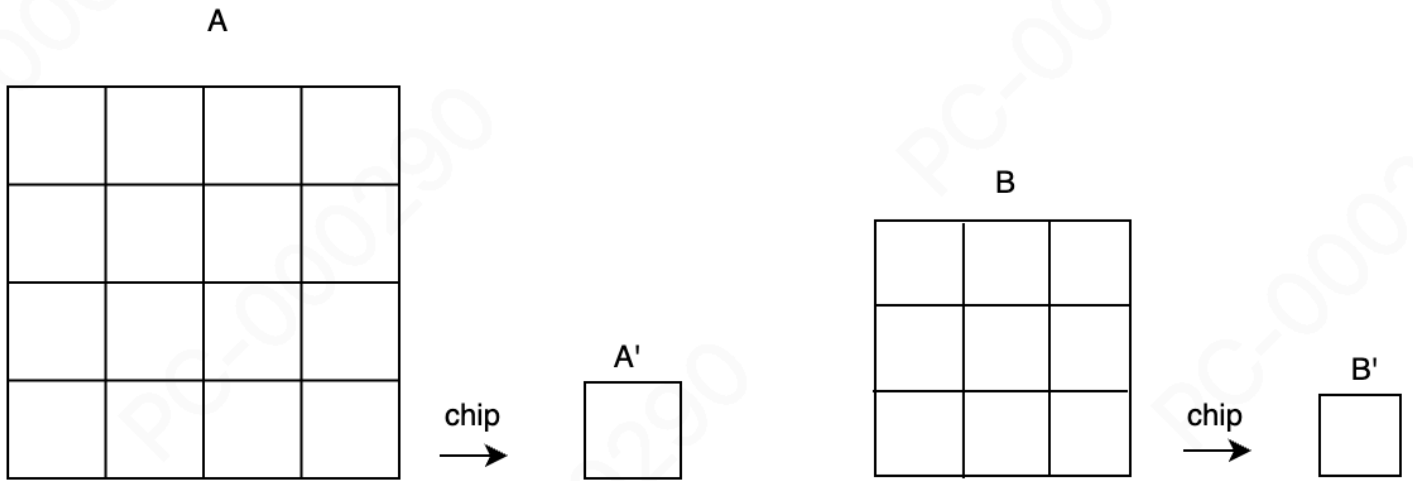
### 3.6 The relationship between transaction size and price of tokens in the market

User transactions cause changes in the positions of any two tokens, which in turn affect the positions of these two tokens relative to other tokens, resulting in synchronized price changes.

| Transaction size | price change   |
|------------------|----------------|
| 10               | 0.000000200000 |
| 50               | 0.000001000000 |
| 100              | 0.000002000002 |
| 500              | 0.000010000050 |
| 1000             | 0.000020000200 |
| 5000             | 0.000100005000 |
| 10000            | 0.000200020002 |
| 50000            | 0.001000500250 |
| 100000           | 0.002002002002 |
| 500000           | 0.010050251256 |
| 1000000          | 0.020202020202 |
| 5000000          | 0.105263157895 |

## 3.7 No impermanent threshold

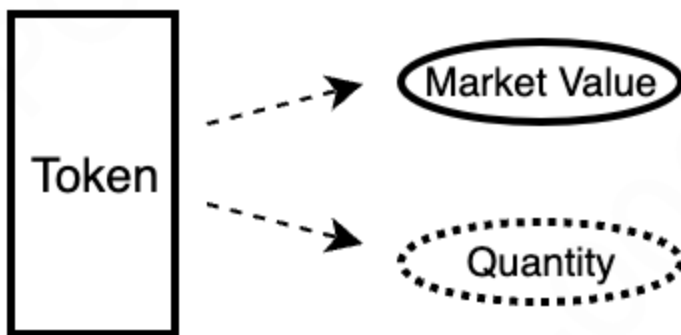
To prevent the platform's tokens from being squeezed out by user transactions, each token is assigned a segmentation number during initialization. Each unit size corresponds to the non-slippage threshold of the token. Therefore, when users transact, if the transaction value is smaller than the non-slippage threshold of the token, there is no impermanent loss. If the transaction exceeds the non-slippage threshold of the token, the transaction will be split into units based on the threshold for execution.



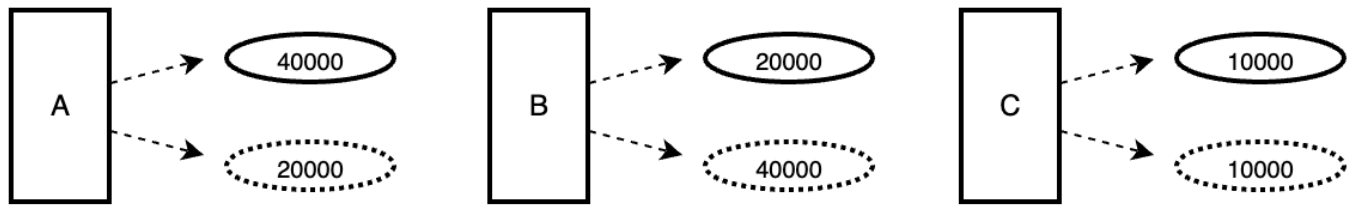
## 4 token

### 4.1 token introduce

Description of the token: The platform possesses 15 units of token A with a market value of 3000. Thus, the token has two attributes: market value and quantity. See the diagram below.



- Noun explanations:  
 Market value: Records the true market value of tokens in the market. When users purchase tokens, the market value of the token increases. When users sell tokens, the market value of the token decreases.  
 Quantity: Records the current quantity of tokens in the market.
- This can be described for any other token as shown in the diagram below.



## 4.2 token type

| type        | introduce                                                                 | Does the transaction incur any fees? | Can invest self alone | Can invest self alone with value token |
|-------------|---------------------------------------------------------------------------|--------------------------------------|-----------------------|----------------------------------------|
| metatoken   | first token in market                                                     | yes                                  | yes                   | no                                     |
| valuetoken  | The product is recognized by the market, with a token ecosystem and team. | yes                                  | yes                   | no                                     |
| normaltoken | Adding new items, Market value to be confirmed                            | yes                                  | No                    | yes                                    |

## 4.3 token Config

- The item configuration occupies 255 positions.

### 4.3.1 Market set

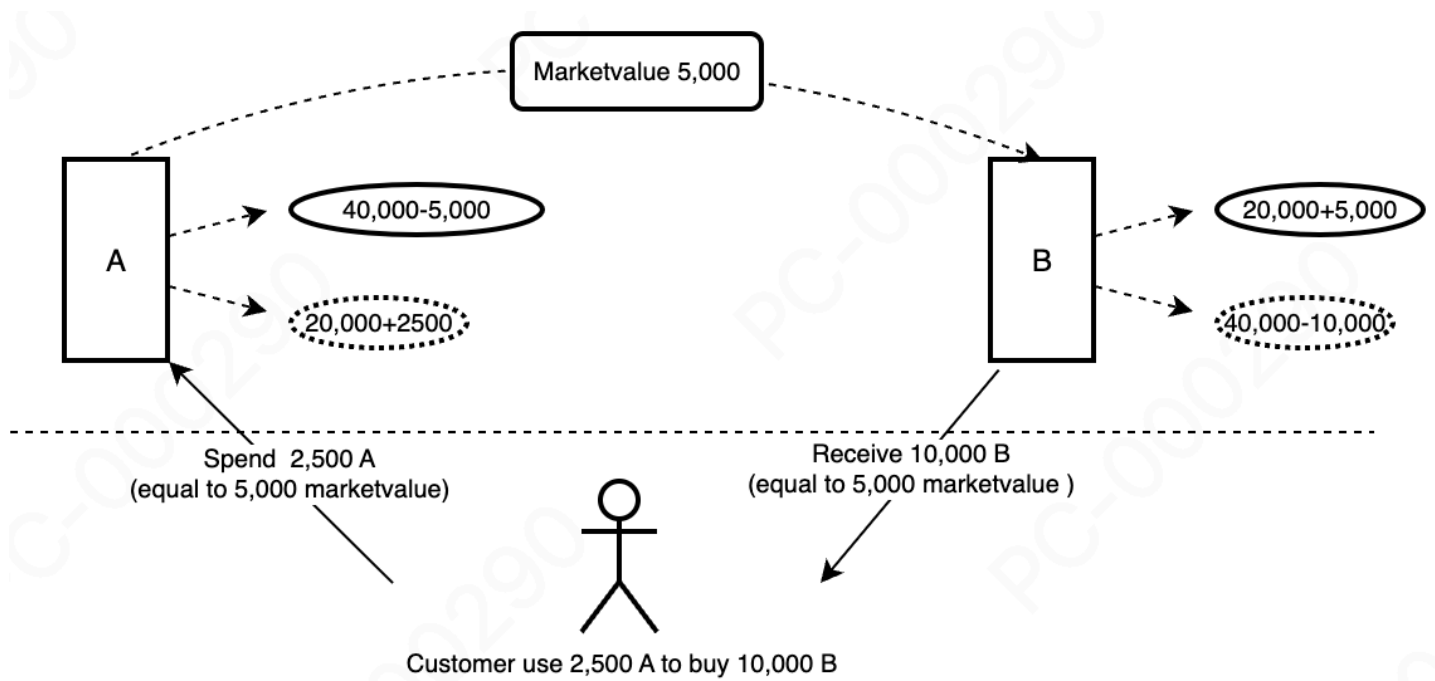
| id  | config     | size | unit    | max | min | start | end | note |
|-----|------------|------|---------|-----|-----|-------|-----|------|
| 1   | valuetoken | 1    | BOOLEAN | 1   | 0   | 1     | 1   |      |
| 2   | recerve    | 32   | 1       | 1   | 0   | 2     | 33  |      |
| ... |            |      |         |     |     |       |     |      |

### 4.3.2 token seller set

| id | config             | size | unit               | max  | min | start | end | note          |
|----|--------------------|------|--------------------|------|-----|-------|-----|---------------|
| 1  | invest fee rate    | 6    | One ten-thousandth | 63   | 0   | 34    | 39  | (0~63)/10000  |
| 2  | disinvest fee rate | 6    | One ten-thousandth | 63   | 0   | 40    | 45  | (0~63)/10000  |
| 3  | buy fee rate       | 7    | One ten-thousandth | 127  | 0   | 46    | 52  | (0~127)/10000 |
| 4  | sell fee rate      | 7    | One ten-thousandth | 127  | 0   | 53    | 59  | (0~127)/10000 |
| 5  | trade chips        | 10   | 64                 | 1023 | 0   | 60    | 69  | (0~1023)X64   |
| 6  | disinvest chips    | 10   | 64                 | 1023 | 0   | 70    | 79  | (0~1023)      |
| 7  | asset type         | 33   | 1                  | ~    | 0   | 80    | 112 |               |
| 8  | telephone          | 48   | 1                  | ~    | 0   | 113   | 160 |               |
| 9  | longitude          | 48   | 1                  | ~    | 0   | 161   | 208 |               |
| 10 | latitude           | 48   | 1                  | ~    | 0   | 109   | 256 |               |

## 5 Swap token

The essence of token exchange is essentially when users exchange token A in the market for token B. By giving up token A, users demonstrate a decrease in the market value of token A. Users only abandon token A when its market value declines. Conversely, users purchase token B when its market value increases.



- As shown in the diagram, when users abandon token A, it leads to an increase in the quantity of token A in the platform and a decrease in its market value. Meanwhile, users acquire token B, resulting in a decrease in the quantity of token B in the platform and an increase in its market value. Consequently, the price relative to token A and token B decreases. In subsequent transactions, the same quantity of token A can only purchase a slightly smaller quantity of token B compared to the previous transaction.
- 

As shown in the diagram, we also adhere to the three fundamental principles of market value conservation in trading:

1. The market value of the tokens used by users during purchase equals the market value of the tokens they acquire.
2. The total market value of the tokens held by users before purchase equals the total market value of all tokens they hold after purchase.
3. User purchases and sales only result in the transfer of market value from one token to another; it does not disappear.

## 5.1 Compute Logical

$$\frac{V_A}{Q_A} * \Delta A = \frac{V_B}{Q_B} * \Delta B \quad (20)$$

$$\text{before trading : } P_{AB} = \frac{V_A * Q_B}{Q_A * V_B} = 4 \quad (21)$$

$$\Delta B = \frac{V_A * \Delta A * Q_B}{Q_A * V_B} = \frac{40000 * 2500 * 40000}{20000 * 20000} = 10000 \quad (22)$$

$$V_A = V_A - \frac{V_A}{Q_A} * \Delta A = 40000 - \frac{40000}{20000} * 2500 = 35000 \quad (23)$$

$$Q_A = Q_A + \Delta A = 20000 + 2500 = 22500 \quad (24)$$

$$V_B = V_B + \frac{V_B}{Q_B} * \Delta B = 20000 + \frac{20000}{40000} * 10000 = 25000 \quad (25)$$

$$Q_B = Q_B + \Delta B = 40000 - 10000 = 30000 \quad (26)$$

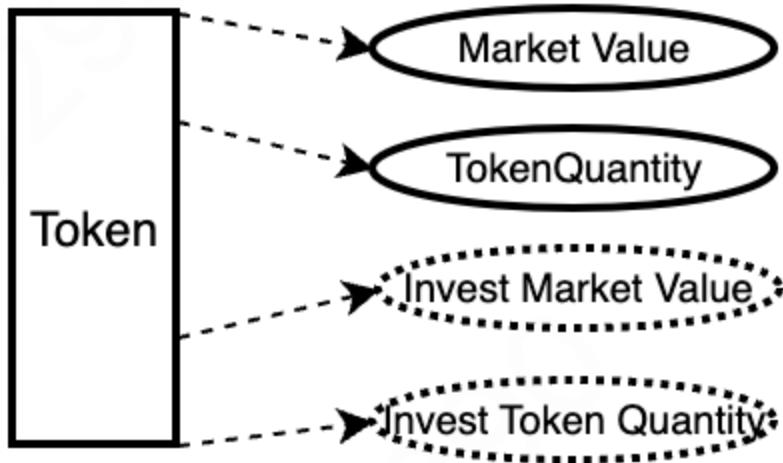
$$\text{after trading : } P_{AB} = \frac{V_A * Q_B}{Q_A * V_B} = 1.86666 \quad (27)$$

Actual compute logical contain the tokens chips ,detail in the contract programing file

## 6 Invest or disinvest token

### 6.1 Record invest data

In market token trading, liquidity must be provided by someone. It is necessary to record the total market value of token investments and the total quantity of investments.

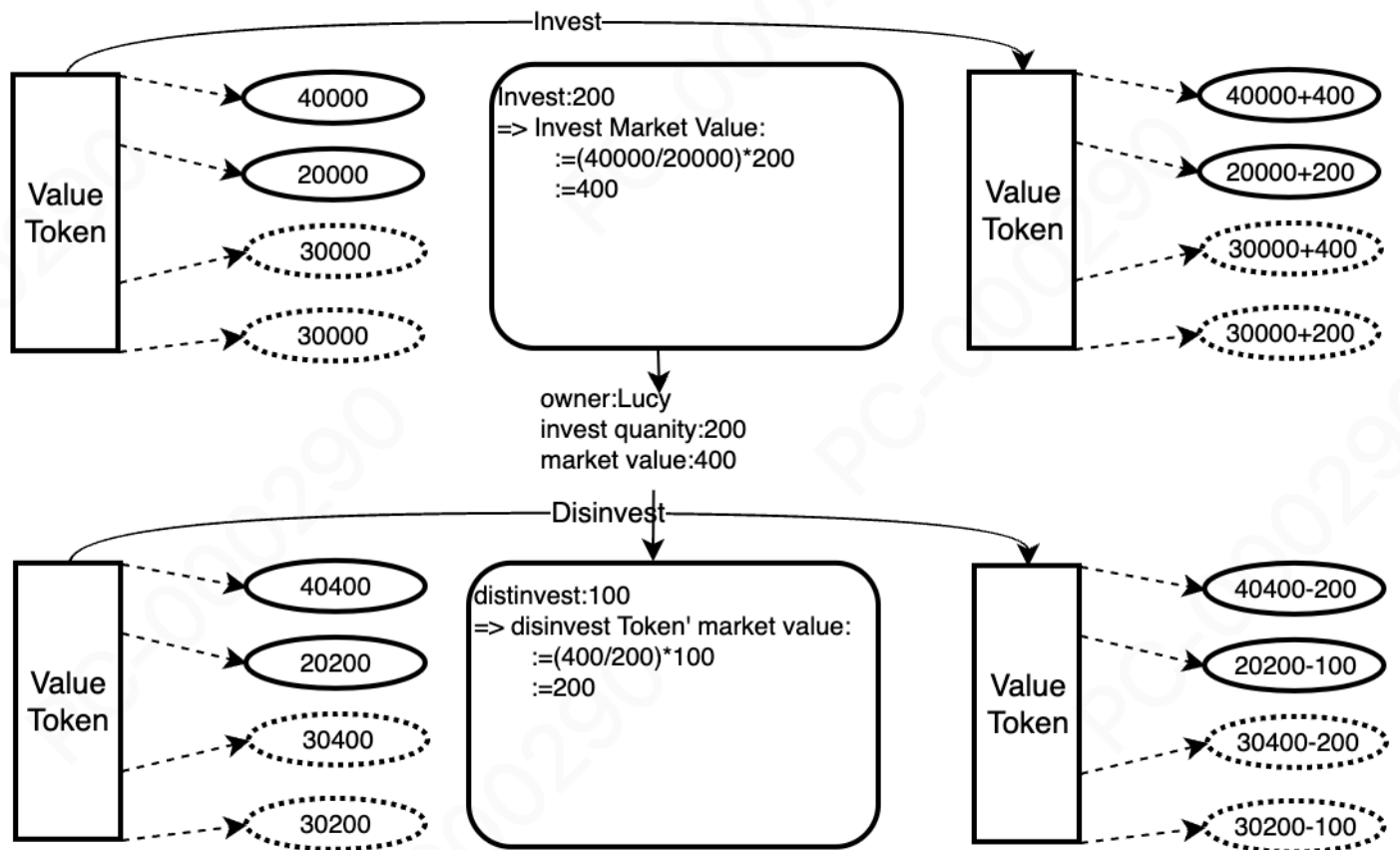


- Noun explanations:

Investment value: Records the total market value of tokens when users invest.

Investment quantity: Records the total quantity of tokens invested by users.

## 6.2 Invest or disinvest value token



- User Invests in Valued tokens:

Users calculate the market value corresponding to the investment quantity based on the current

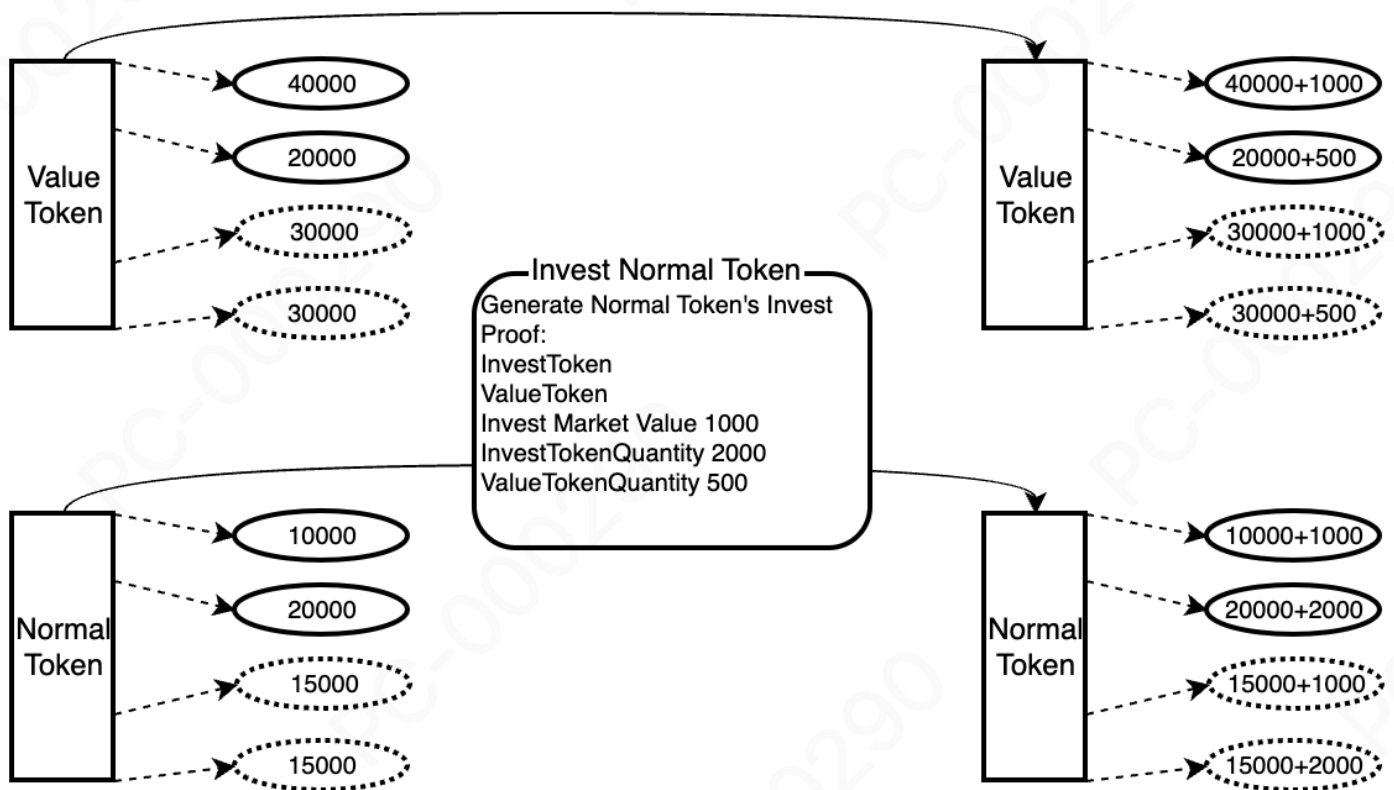
status of valued tokens. This facilitates profit calculation when withdrawing investments.

- User Withdraws from Valued tokens:

Users calculate the profits generated from investments based on investment records.

When withdrawing from tokens, the canceled quantity or the canceled market value corresponding to the quantity needs to be less than the total current quantity or the total value divided by the maximum withdrawal ratio.

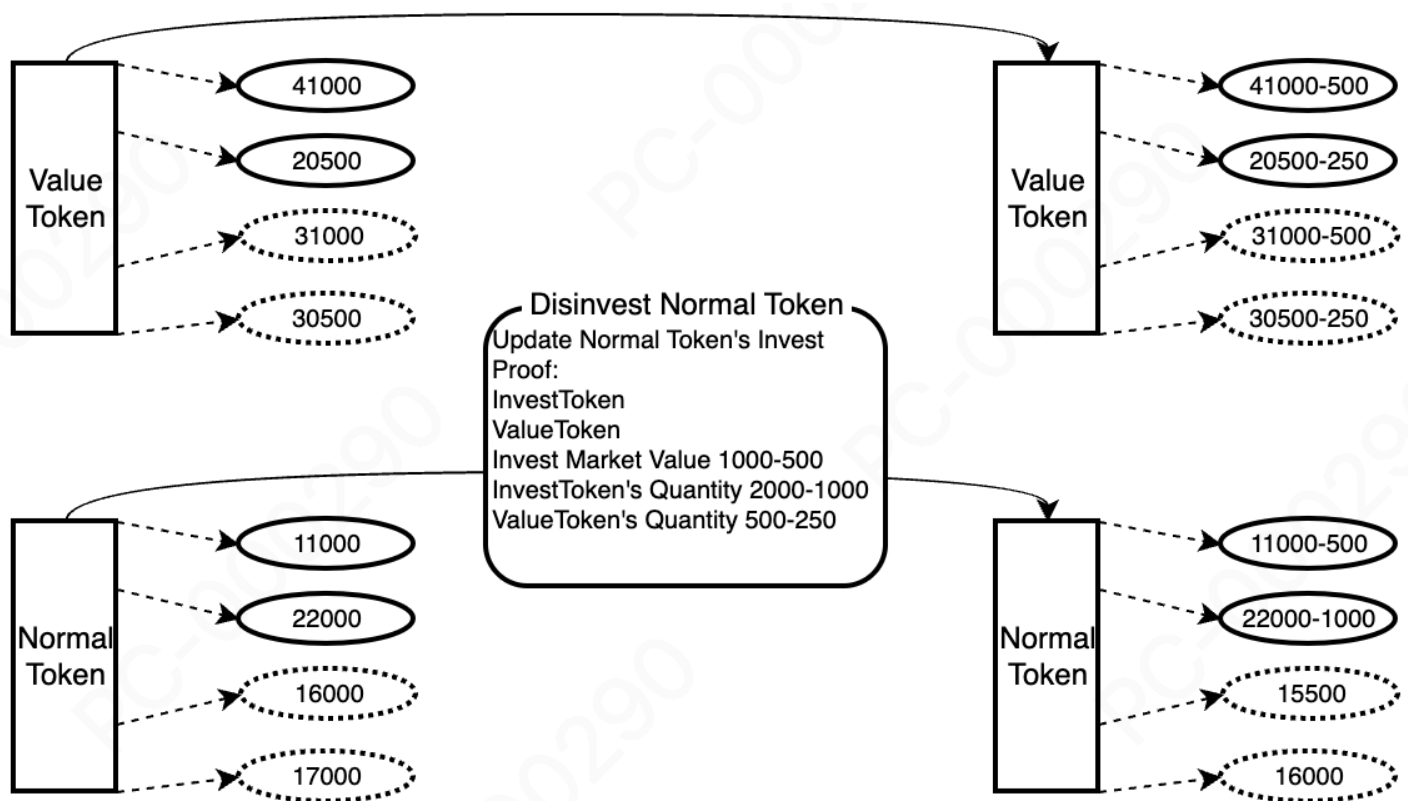
## 6.3 Invest normal token



- User Invests in Regular tokens:

Due to the volatile market value of regular tokens, it is easy to form arbitrage against other users' tokens on the platform. To avoid this situation, it is necessary to invest in valued tokens with comparable market values. Both valued tokens and regular tokens generate investment returns, as detailed in the fee distribution.

## 6.4 Disinvest normal token

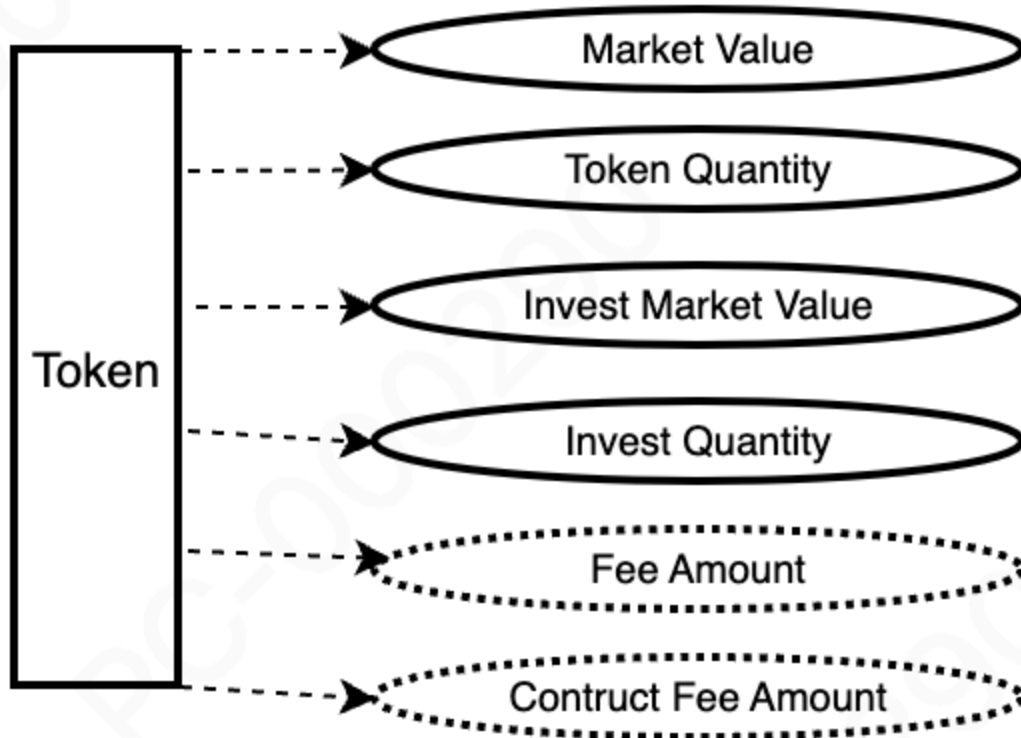


- User Invests in Regular tokens:

Due to the volatile market value of regular tokens, it is easy to form arbitrage against other users' tokens on the platform. To avoid this situation, it is necessary to invest in valued tokens with comparable market values. Both valued tokens and regular tokens generate investment returns, as detailed in the fee distribution.

# 7 token's fee

## 7.1 Record token's fee

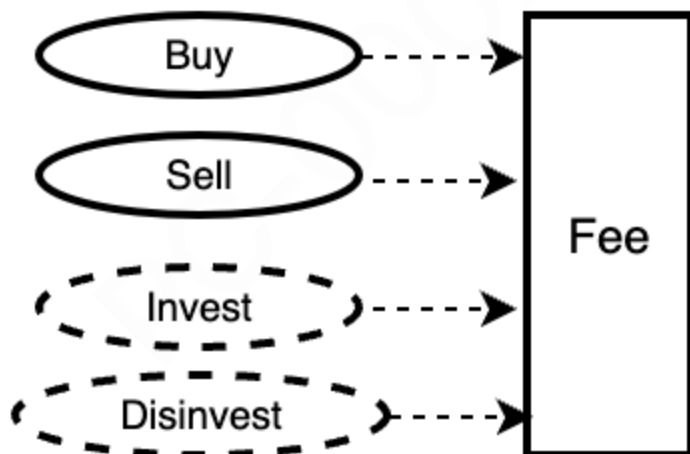


- Noun explanations:

Total fees refer to the sum of actual transaction fees generated and construction fees.

Construction fees are virtual fees introduced to calculate the profits generated from user investments, but they are not actual transaction fees. For more details, please refer to Sections 7.4 and 7.5.

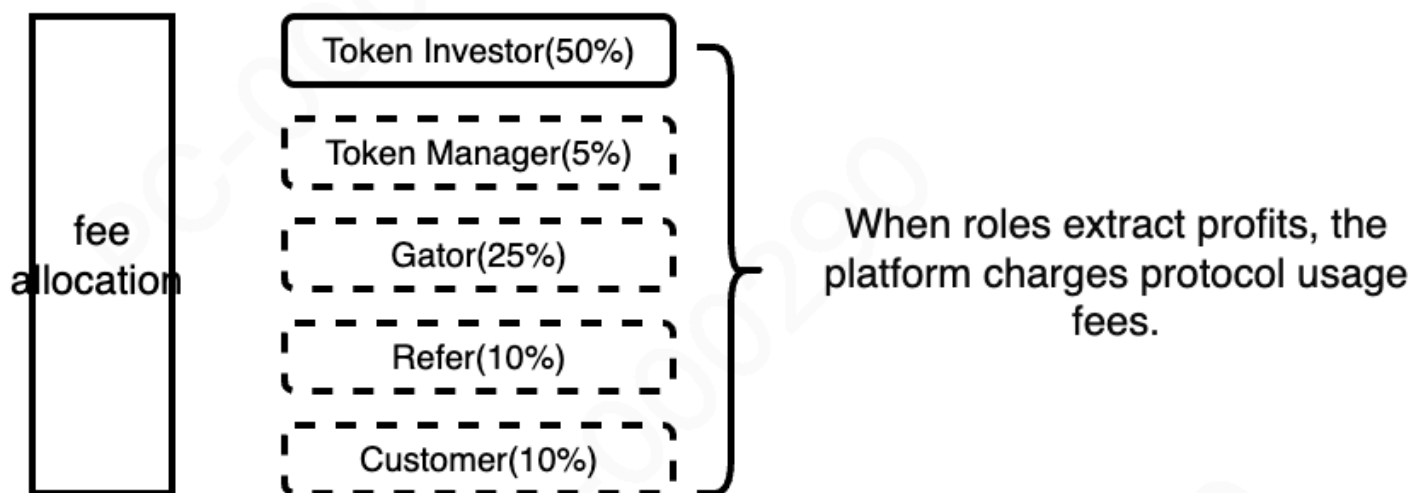
## 7.2 Fee source



The fee ratio will be adjusted by Token owner based on the actual situation.

The source of transaction fees (actual transaction fees) is calculated based on the fee rate of tokens when users perform operations.

## 7.3 Fee allocate



When roles extract profits, the platform charges protocol usage fees.

The allocation ratio will be continuously adjusted based on the actual situation.

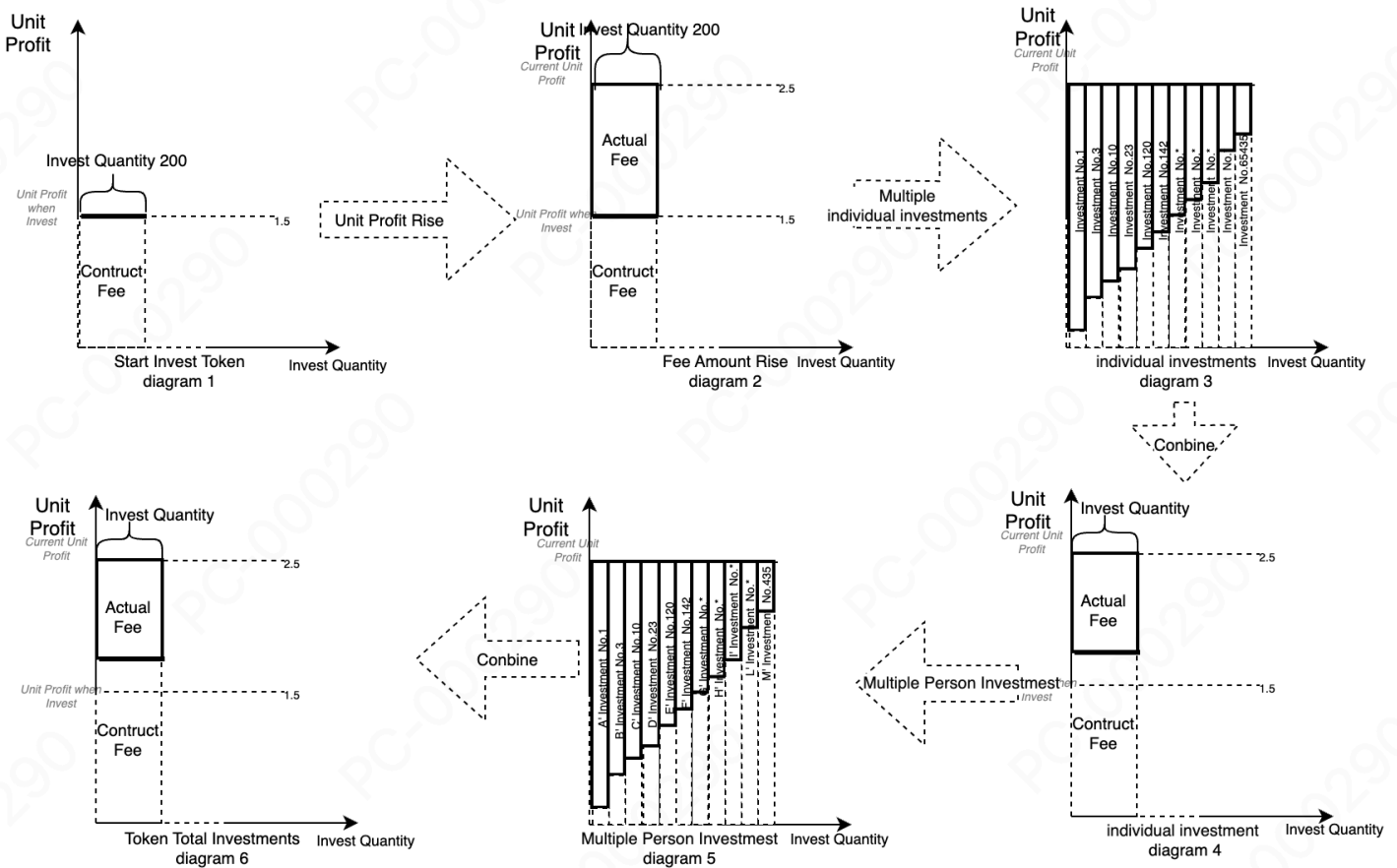
The platform involves platform technology, portal operation, referrers, users, and liquidity providers.

The platform will distribute profits reasonably.

The fee distribution for liquidity providers can be found in Section 7.4, Fee Process.

- If users fill in a referrer:  
The allocation for each role is recorded in real-time.
- If users do not fill in a referrer:  
The proportion held by users is allocated to merchants.  
The proportion held by referrers is allocated to the portal.

## 7.4 Fee compute flow(invest)



- diagram 1  
Unit fee refers to how much fee each unit of investment should receive, calculated as the total fee amount divided by the total investment quantity. As transactions progress, fees continuously generate, leading to an increase in the total fee amount, and consequently, an increase in the unit fee.  
The construction fee is introduced at the beginning of user investment to record the total fee

amount that users should not enjoy. It is calculated as the investment quantity multiplied by the unit fee at the time of investment.

- diagram 2

When fees continue to generate within the platform, the unit fee will continuously increase.

The profit generated from user investments is calculated as follows: Profit = (Unit fee X Investment quantity) - Construction fee.

- diagram 3

When users make multiple investments in the same token, they can consolidate them into a single investment record.

The consolidated construction fee after merging equals the sum of the construction fees before merging.

The profit generated from user investments is calculated as follows: Profit = (Unit fee X Investment quantity) - Consolidated construction fee.

- diagram 4

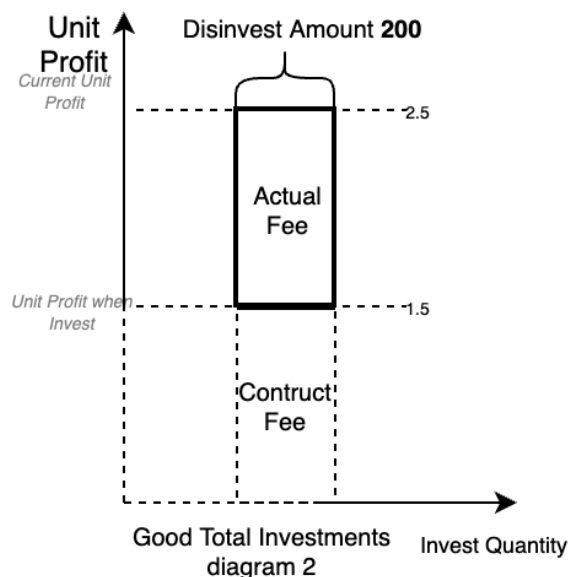
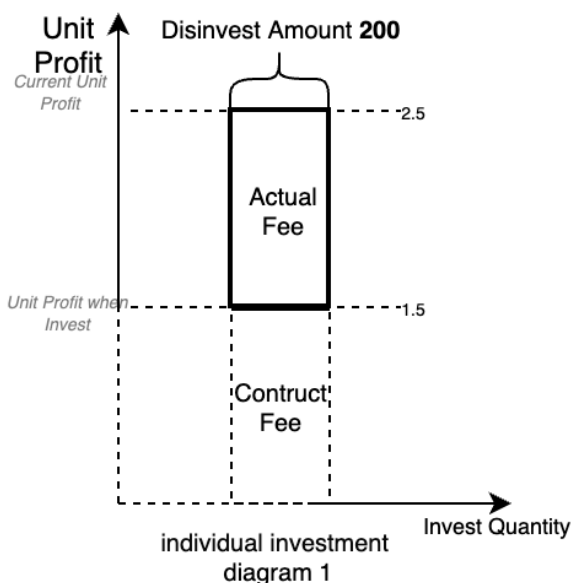
The diagram illustrates the consolidated investment situation.

- diagram 5

When multiple users invest, it can be aggregated into the total investment quantity, total investment market value, and total construction fee for this token.

The total actual investment profit for this token at present equals the current total fees minus the aggregated construction fees.

## 7.5 Fee compute flow(disinvest)



- diagram 1

When users withdraw their investments, the profit gained equals (Unit fee X withdrawal quantity) - (Construction fee X (withdrawal quantity / total investment quantity)).

- diagram 2

The profit and construction fee incurred when subtracting user withdrawals from the token.

## 8 Market config

| id  | config         | size | unit        | max | min | start | end | note |
|-----|----------------|------|-------------|-----|-----|-------|-----|------|
| 1   | token investor | 6    | One percent | 63  | 0   | 256   | 251 |      |
| 2   | token manager  | 6    | One percent | 63  | 0   | 250   | 245 |      |
| 3   | gater          | 6    | One percent | 63  | 0   | 244   | 239 |      |
| 4   | referer        | 6    | One percent | 63  | 0   | 238   | 233 |      |
| 5   | customer       | 6    | One percent | 63  | 0   | 232   | 227 |      |
| 6   | plat           | 6    | One percent | 63  | 0   | 226   | 221 |      |
| ... |                |      |             |     |     |       |     |      |

## 9 Main code implementation (see code for details)

### 9.1 Deploy Contract gas

| Deployment Cost | Deployment Size |
|-----------------|-----------------|
| 5144500         | 25351           |

## 9.2 Function(main function)GAS

| Function Name  | first  | second | third | note                               |
|----------------|--------|--------|-------|------------------------------------|
| buytoken       | 77728  | 42128  | 42128 |                                    |
| disinvestProof | 153111 | 53811  | 53811 | devest normal token                |
| disinvestProof | 100438 | 40438  | 40438 | devest value token                 |
| inittoken      | 406979 |        |       | init normal token                  |
| investtoken    | 291576 | 48099  | 48099 | invest normal token                |
| investtoken    | 216564 | 36187  | 36187 | invest value token                 |
| collectProof   | 209386 | 42486  | 42486 | collect normal token invest profit |
| collectProof   | 125979 | 38479  | 38479 | collect value token invest profit  |
| tokenwarefare  | 54688  | 33588  | 27988 | token walefare                     |

at same token on same action,the gas cost

## 10 Legal License

### 10.1 Description

To uphold the proper rights of the project and facilitate understanding of the agreements by other users, different files are governed by different open-source licenses. Violations of these agreements may result in legal consequences.

### 10.2 Protocol Description

Documents using the MIT license are freely available for everyone to use.

Documents using the BUSL-1.1 license can only be used for learning purposes within the term of the agreement and cannot be used for commercial purposes. For specific terms of the agreement, please refer to the LICENSE file in the project or on GitHub: <https://github.com/ttswap/ttswap->

[core/LICENSE](#). If the project inadvertently violates other open-source licenses, please contact us immediately, and we will make adjustments promptly.

## 11.3 Open Source License Information

### Contract

- |—— TTSwap\_Market.sol(BUSL-1.1)
- |—— TTSwap\_NFT.sol(BUSL-1.1)
- |—— TTSwap\_Token.sol(BUSL-1.1)
- |—— TTSwap\_LimitOrder.sol(BUSL-1.1)
- |—— interfaces
  - | |—— I\_TTSwap\_Market.sol(MIT)
  - | |—— I\_TTSwap\_Token.sol(MIT)
  - | |—— I\_TTSwap\_LimitOrderMaker.sol(MIT)
  - | |—— I\_TTSwap\_LimitOrderTaker.sol(MIT)
  - | |—— I\_TTSwap\_NFT.sol(MIT)
  - | |—— I\_TTSwap\_APP.sol(MIT)
- |—— libraries
  - |—— L\_Good.sol(BUSL-1.1)
  - |—— L\_GoodConfig.sol(MIT)
  - |—— L\_MarketConfig.sol(MIT)
  - |—— L\_Proof.sol(BUSL-1.1)
  - |—— L\_TTSwapUINT256.sol (MIT)
  - |—— L\_Currency.sol (MIT)
  - |—— L\_Struct.sol (MIT)
  - |—— L\_OrderStatus.sol (MIT)
  - |—— L\_TakeLimitPriceOrder.sol (MIT)
  - |—— L\_Lock.sol (MIT)
  - |—— L\_TTSTokenConfig.sol(MIT)

### docs

- |—— ebook
- |—— whitepaper-cn
  - |—— whitepaper-cn.pdf(BUSL-1.1)
- |—— whitepaper-en
  - |—— whitepaper-en.pdf(BUSL-1.1)

# 12 contract

X:@ttswap\_exchange

TG:@ttswap01

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